31/01/26

CO 2 :- CORROSION AND IT'S CONTROL

Corrosion :- Distruction of metal by attack of unwanted gas present in the environment.

→ All metals except gold and platinum are available in form of "metalic ore"

The another reason for corrosion is when any metal is extracted from its ore by the process of reduction then pure metal is unstable so immediately it reacts with gas and undergoes corrosion

→ Corrosion takes place only with metals because they have ability to loose e^-

Theories of Corrosion :-

- Dry Corrosion (or) Chemical corrosion
  - corrosion by Oxygen
  - Corrosion by other gas
  - Liquid metal corrosion
- Wet Corrosion (or) Electro-chemical corrosion
  - Evolution of Hydrogen
  - absorbtion of Oxygen

Dry Corrosion :-

Corrosion takes place in absence of moisture

Wet Corrosion :-

Corrosion takes place in presence of moisture.

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Types of dry Corrosion :-

1. Oxidation Corrosion :-
Corrosion of metal by the action of oxygen in absence of moisture and forms metal-oxide layer
⇒ Further action of oxygen depends on the nature of metal-oxide layer formed on the metal.
1. If the layer is stable further corrosion does not takes place.

2. If the layer is unstable then further corrosion takes place.

2. Corrosion by Other gases :-
Corrosion of metal by action of oxygen and other gases in absence of moisture. Further corrosion depends on stability of metal-oxide layer.

3. Liquid-Metal Corrosion :-
Passing of hot liquid metal on to the surface of solid metal then it weakens the strength of solid metal is called Liquid-Metal Corrosion
→ Liquid metal penitrates into solid metal

Types of wet Corrosion :-

1. Evolution of Hydrogen :-
When we place any metal in moist environment then anode and cathode cells are formed :
① when rod is at acidic solution
At anode :-

$$Fe \rightarrow Fe^{+2} + 2e^-$$

At cathode :- $2H^+ + 2e^- \rightarrow H_2 \uparrow$

H^+ ions are released from cathode and corrosion takes place at anode.

② When rod is at Basic medium
$Fe(OH)_3$ is formed.
→ Anode and cathode are developed on metal.
OH^- reacts directly with anode.

③ When rode is at neutral medium
Both O_2 and H_2O are present.

② Absorption of Oxygen :-
Both H_2O and O_2 are present. These two together form OH^- ions ~~are~~ and reacts with Fe^{2+} and forms Fe_2O_3

$$\cancel{2H_2}\ \tfrac{1}{2}O_2 + H_2O \longrightarrow 2OH^-$$

$$Fe^{+2} + 2OH^- \longrightarrow Fe(OH)_2$$

$$Fe(OH)_2 + O_2 + 2H_2O \longrightarrow 4Fe(OH)_3 \text{ (or) } 3Fe_2O_3 + 3H_2O \text{ (rust)}$$

Types of Corrosion - Electro-chemical
1. Galvanic corrosion
2. pitting corrosion
3. Differential aeration corrosion
4. Stress corrosion.

1. Differential aeration Corrosion :-
→ high demand area acts as cathode
→ low demand area acts as anode.

On one single metal two different areas ① painted ② Non-painted.

It is caused by varying electrolyte concentration or aeration levels.

When a poorly oxygenated area is adjacent to an area with a good supply of oxygen an anodic / cathodic reaction occurs.

→ Oxygen-less metal parts acts as anodes while oxygen-rich metal parts acts as cathode.

Reaction at anode :- $Zn \rightarrow Zn^{+2} + 2e^-$

Reaction at cathode :- $1/2O_2 + H_2O + 2e^- \rightarrow 2OH^-$

Stress - Corrosion

It occurs when a relatively inert material is subjected to a significant amount of stress.

→ Stress can either be externally applied or residually present.

→ Metal develops internal stress during manufacturing processes like fabrication, heat treatment and rolling.

→ Areas under stress act as anodes having high energy levels & electrode potential

→ Areas that are stress-free act as cathode and have low electrode potentials and low energy levels.

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Ex :-

1. Caustic Embrittlement :-
   Caused by excessive NaOH in boilers

2. Underground ~~and~~ steel pipelines

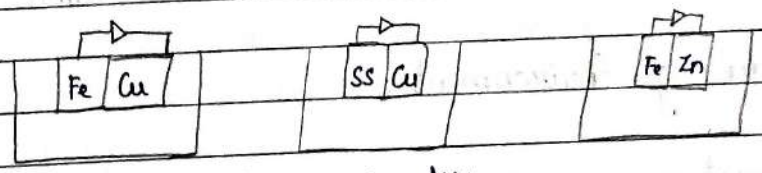
3. Galvanic or Bimetallic Corrosion

→ It is an electrochemical process in which one metal corrodes before another metal is in contact with through an electrolyte.

→ When two dissimilar metals are electrically connected immersed and exposed to a neutral environment in presence of O_2

→ Metals with lower reduction potential spontaneously oxidize and corrosion

→ Anode attack rates accelerates with time.



It requires three Conditions
1. Dissimilar metal

2. Electrical ~~corrosion~~ contact

3. Moisture environment.

Factors influencing Corrosion

1. Nature of metal position of metal in galvanic series	Nature of Environment Temperature
2. Over Voltage	pH
3. purity of metal	Humidity
4. Nature of corrosion product	Effect of Oxidation and Oxidizers
5. Nature of oxide film	Presence of impurities in atmosphere
	Nature of ions present
	Formation of Oxygen Concentration cell.

I Nature of Environment :-

① Temperature :-

The rate of corrosion reaction and diffusion rate increases with increase in temperature, causing the increase in rate of corrosion.

② pH :-

pH has greater effect on corrosion. When pH is lowered the rate of corrosion increases

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③ Humidity :-

Corrosion of metals becomes faster in humid atmosphere. It provides water, which is essential for setting up an electrochemical corrosion.

$R_{oc} \propto$ humidity

④ Effect of Oxidation and Oxidizers :-

When oxidizer is added, some metals achieve passivity due to formation of oxide film which is protective in nature.

⑤ presence of impurities in atmosphere :-

In presence of impurities in atmosphere, the acidity adjacent to the metal surface increases and its electrical conductivity also increases.

This results in increase of corrosion.

⑥ Nature of ions present

- presence of anions in the medium leads to the formation of insoluble reactions products which inhibit further corrosion
- If chloride ions present in medium destroy the protective and passive surface film thereby exposing the metal/alloy the surface for fresh corrosion.
- Many metals undergo corrosion rapidly, if corroding medium contains ammonium salts
- presence of even traces of copper in mine water accelerates the corrosion of iron pipes.

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7. Formation of Oxygen concentration cell
With the increase in the supply of oxygen to the moist metal surface the corrosion is promoted.

Nature of Metal

① position of metal in galvanic series:-
Metal present above galvanic series elements acts as anode. Below the galvanic series acts as cathode.

② Over Voltage :-

$$\text{Over voltage} \propto \frac{1}{\text{rate of corrosion}}$$

③ purity of metal :-

$$\text{purity of metal} \propto \frac{1}{\text{rate of corrosion}}$$

④ Nature of corrosion product :-
→ If metalic-oxide is volatile and soluble the corrosion is faster

→ If metalic-oxide is insoluble the corrosion does not takes place.

⑤ Nature of Oxide film :-
ratio of oxide film is more then corrosion is less
if ratio of oxide film is less then corrosion is more.

Pilling - Bed Worth rule :

$$= \frac{\text{Vol. of the metal oxide formed}}{\text{Vol. of the metal consumed}}$$

if :-

ratio < 1 :- Non-protective layer

ratio >= 1 :- protective layer.

Relative area of electrode

rate of corrosion ∝ $\frac{\text{area of electrode}}{\text{area of anode}}$

Cathodic protection

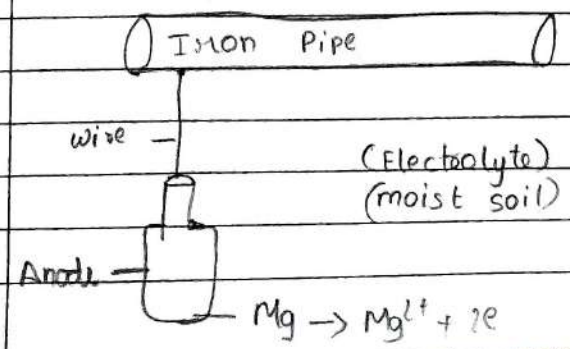
The metal to be protected is forced to act as cathode so that the metal can be protected from corrosion.

There are two types of Cathodic protection

① Sacrificial Anode protection

② Impressed Current Cathodic protection

① Sacrificial Anodic protection :-



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Imp applications are protection of buried lines underground cables, marine structures

2. Impressed Current Cathodic protection

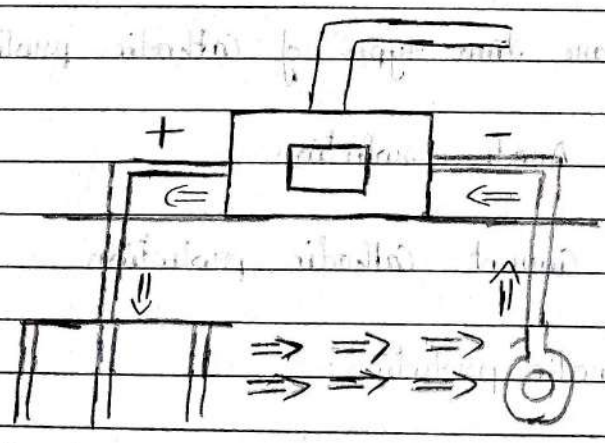
- Using this method, the metallic object to be protected is made cathode by connecting it to the negative terminal of DC source.

- Positive terminal is connected to an insoluble anode such as graphite, scrap iron (or) platinum.

Applications :-

- Submarines are protected against corrosion by connecting the negative terminal of DC generators to surface.

- Water tanks, buried oil (or) water pipelines, transmission line towers are protected by this method.



Metallic Coatings :-

Coating of metal on the base metal is called metallic coating.

It is divided into two types :-

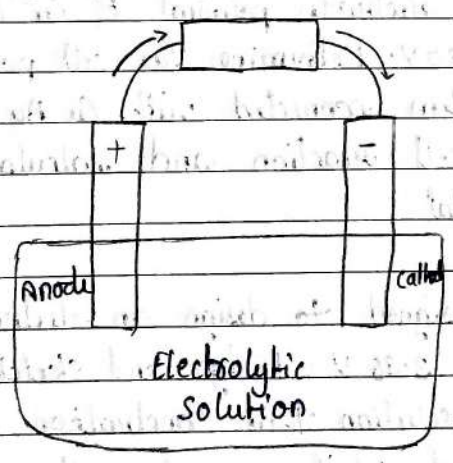
① Anodic Coating

② Cathodic Coating

Electroplating :-

Coating of coat metal on the surface of base metal in presence of electric current is called electroplating

Electroplating of copper :-



At anode :-

$$Cu \longrightarrow Cu^{+2} + 2e^-$$

At cathode :-

$$Cu^{+2} + 2e^- \longrightarrow Cu$$

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Electroplating of Nickel :-

$NiCl_2$ or Nickel sulphate is used.

At anode :-

$$Ni \longrightarrow Ni^{+2} + 2e^-$$

At cathode :-

$$Ni^{+2} \longrightarrow Ni$$

Problems :-

1. Draw a fully labeled diagram showing how to measure e° value for Mg^{+2}/Mg electrode and Ag^+/Ag electrode

$$E^\circ (Ag^{+}/Ag) = 0.80V$$
$$E^\circ (Mg^{+2}/Mg) = -2.37V$$

2. Given the Standard electrode potential $E^\circ\ Cu^{+2}/Cu = 0.34V$ and $E^\circ\ Ag^+/Ag = 0.80V$. Determine the cell potential for galvanic cell when connected with Cu Ag cell. Write and draw cell reaction and calculate standard cell potential

3. An Engineer is assigned to design an electro chemical cell that will deliver 3.36 V. Design and sketch a cell voltage, detailing the solution their concentration and electrode you will need. Write eqns for all reactions

1. $E^\circ_{cell} = E_c - E_a$

$E^\circ_{cell} = 0.80 - (-2.37)$

$= 3.17$

2. Given :-

$E^\circ_{Cu^+/Cu} = 0.34\,V$

$E^\circ_{Ag^+/Ag} = 0.84\,V$

At anode :- $Cu \rightarrow Cu^{+2} + 2e^-$

At cathode :- $Ag^{+2} \rightarrow Ag + 2e^-$

$Cu + 2Ag \longrightarrow Cu^{+2} + 2Ag$.

3. $E_{cell} = E_c - E_a$

$3.85\,V = E_{Ag^+/Ag} - E_{Li^-/Li}$

$= \cancel{0.80} \; 0.80 - (-3.05)$

$= 3.85\,V$

Anode : Li/Li^{+2} cathode : Ag^+/Ag